

Stormwater Regulation Summary: Virginia (Statewide)

Municipal Stormwater Management Technical Profile | Sempergreen USA

1. Regulatory Overview

- Primary driver: Chesapeake Bay TMDL — Virginia has the largest pollutant reduction obligation of any Bay state.
- Virginia Erosion and Stormwater Management Program (VESMP, renamed from VSMP effective July 2024) administered by VA DEQ.
- Unique approach: Phosphorus-based design — BMPs are credited based on phosphorus removal, not just volume.
- Applies to ≥ 1 acre land disturbance (VESMP) or $\geq 2,500$ SF in Chesapeake Bay Preservation Areas (CBPA).
- Local MS4 permittees enforce VESMP requirements — Northern VA, Hampton Roads, and Richmond areas are most active.

2. Typical Soil Conditions

- Virginia spans three physiographic provinces with very different soil conditions:
- Coastal Plain (east): Sandy loam to sandy clay loam; HSG A-B; good infiltration potential.
- Piedmont (central): Residual clay soils from metamorphic/igneous rock; HSG C-D; poor infiltration.
- Valley and Ridge / Blue Ridge (west): Variable — residual clays in valleys (C-D), rocky soils on ridges.
- Northern VA (Fairfax, Arlington, Loudoun) is predominantly Piedmont clay — HSG C-D.
- Implication: Infiltration feasibility varies dramatically by region — Coastal Plain is favorable, Piedmont is challenging.

3. Green Roof Requirements

- No statewide green roof mandate.
- Green roofs receive favorable phosphorus removal credits in the VA BMP Clearinghouse — making them effective for VESMP compliance.
- VA DEQ BMP Clearinghouse assigns pollutant removal efficiencies to each approved BMP, including green roofs.
- Northern VA jurisdictions (Arlington, Fairfax, Alexandria) have higher density and site constraints that favor rooftop BMPs.
- VA Clean Economy Act and PACE financing support combined green roof + solar installations.
- Data center corridor in Northern VA presents a large-footprint opportunity — massive flat roofs with corporate ESG commitments.

4. TSS Requirements

- Virginia's VESMP uses phosphorus as the primary pollutant design metric — not TSS directly.
- However, BMPs that achieve phosphorus reduction typically also achieve significant TSS removal.
- *VA BMP Clearinghouse provides pollutant removal rates for each approved BMP — verify current green roof TP removal credit in DEQ BMP Clearinghouse (legacy site migrated to DEQ website July 2024).*
- The phosphorus-based approach means green roofs may receive higher compliance credit than in TSS-only jurisdictions.
- *Phosphorus removal credit percentages vary by BMP design — verify current values in VA BMP Clearinghouse (2024 update).*

5. Nature-Based Retention Requirements

- Phosphorus-based: New development must achieve a specified phosphorus load reduction based on land use and receiving waters.
- The required TP load is calculated per VESMP regulations — BMPs are selected to achieve the target load reduction.
- CN (Curve Number) method is used for volume and peak flow calculations — standard NRCS TR-55 approach.
- Green roofs are credited in the VA BMP Clearinghouse for both volume reduction and pollutant removal.
- Typical retention target approximately equivalent to 1.0" WQv in most jurisdictions.
- Chesapeake Bay Preservation Areas (CBPA) have additional riparian buffer and impervious cover requirements.

6. Allowable Outflow Rates

- Channel protection (Cpv): 24-hour extended detention of the 1-year, 24-hour storm event.
- Flood protection: 10-year storm peak control to pre-development rate.
- Virginia does not specify a flat l/s/ha rate — control is relative to pre-development conditions.
- For reference, typical pre-development rates for Piedmont sites (CN 65-70): approximately 15-30 l/s/ha for the 10-year storm.
- *Local MS4 permittees may impose stricter flow control — verify with specific jurisdiction.*
- Some Northern VA localities require 2-yr through 100-yr peak control.

Items in *italic* should be verified against current municipal codes. | Generated by Sempergreen USA Stormwater BMP Comparison Tool